ease write your Enrollment Number) Enrollment No: 015

# End-Term Examination (REGULAR)
## (CBCS)(SUBJECTIVE TYPE)(Offline)
## Course Name:<B.Tech. - IT/MAE/DMAM> Semester:<2nd>
## JULY 2023

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| ubject Code: BAS 102 | Subject: Applied Mathematics II | | |
| ime :3 Hours | | Maximum Marks :60 | |
| lote : Q1 is compulsory. Attempt one question each from the Units I, II, III & IV. | | | |
| Q1 | | | |
| | (a) If $\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$ and $\lvert\vec{r}\rvert = r$, then prove that $grad\left(\frac{1}{r}\right) = -\frac{\vec{r}}{r^3}$. | (2.5*8=20) | |
| | (b) Describe the physical interpretation of curl. | | |
| | (c) Solve $(D^2 - 2D + 4)^2 y = 0\,;\, D = \frac{d}{dx}$. | | |
| | (d) Define the term "Wronskian" of the solutions of a homogeneous differential equation. Check whether the functions 1, $e^{-x}$ and $e^{-2x}$ are linearly independent or not using Wronskian. | | |
| | (e) Find the Laplace transform of $\int_0^t \frac{\sin x}{x}\,dx$. | | |
| | (f) Find the Laplace transform of $f(t) = \begin{cases} \cos t, & 0 < t < \pi \\ 0, & t \geq \pi. \end{cases}$ | | |
| | (g) If $u = \frac{1}{2}\log(x^2 + y^2)$, then determine its analytic function. | | |
| | (h) Evaluate $I = \oint_C \frac{dz}{z-2}$ around<br>(i) Circle $\lvert z - 2\rvert = 4$<br>(ii) Triangle with vertices at (0,0), (1,0), (0,1). | | |
| | UNIT-I | | |
| Q2 | (a) Evaluate the line integral of $\vec{F}$ over the circular path given by $x^2 + y^2 = a^2, z = 0$ where $\vec{F} = (\sin y)\,\hat{\imath} + x(1 + \cos y)\hat{\jmath}$.<br><br>(b) Evaluate $\iint \vec{F}.d\vec{S}$ using Gauss divergence theorem where $\vec{F} = 2xy\,\hat{\imath} + yz^2\hat{\jmath} + zx\hat{k}$ and $S$ is the surface of the region bounded by $x = 0, y = 0, z = 0, y = 3, x + 2z = 6$. | (5*2=10) | |
| Q3 | (a) Evaluate $\iint \vec{F}.\hat{n}\,dS$, over $S$ where $\vec{F} = y\,\hat{\imath} + 2x\,\hat{\jmath} - z\,\hat{k}$ and $S$ is the surface of the plane $2x + y = 6$ in the first octant cut off by the plane $z = 4$.<br><br>(b) Verify Green's theorem for<br><br>$\int_C \left(\frac{1}{y}\,dx + \frac{1}{x}\,dy\right)$ | (5*2=10) | |

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| | where $C$ is the boundary of the region bounded by the parabola $y = \sqrt{x}$ and the lines $x = 1,\ x = 4,\ y = 1$. | | |
| | **UNIT-II** | | |
| Q4 | (a) Solve $(D^2 + 5D + 4)y = x^2 + 7x + 9$ by Operator method.<br>(b) Solve $(D^2 - 1)y = e^{-x}\sin(e^{-x}) + \cos(e^{-x})$ by Variation of Parameter method. | (5*2=10) | |
| Q5 | (a) Solve $x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 5y = x^2\sin(\log x)$.<br>(b) Solve $(D^2 - 2D)y = e^x \sin x$ by method of Undetermined Coefficients. | (5*2=10) | |
| | **UNIT-III** | | |
| Q6 | (a) State and prove First Shifting theorem and hence evaluate the Laplace transform of $e^{-t}sin^2\,3t$.<br>(b) Find inverse Laplace transform of $\frac{1}{(s-2)(s+2)^2}$ by using convolution theorem. | (5*2=10) | |
| Q7 | (a) Find the Laplace transform of $(1 + te^{-t})^3$.<br>(b) Solve $(D^2 - 2D + 1)y = e^t$, $y = 2$ and $Dy = -1$ at $t = 0$ by using Laplace transforms. | (5*2=10) | |
| | **UNIT-IV** | | |
| Q8 | (a) Show that for $f(z) = \begin{cases} \frac{2xy(x+iy)}{x^2+y^2}, & if\ z \neq 0 \\ 0, & if\ z = 0 \end{cases}$<br>C-R equations are satisfied at origin but derivative of $f(z)$ at origin does not exist.<br>(b) Find the residue of $f(z) = \frac{z^2}{(z^2+1)^2}$ at all its isolated singular points. | (5*2=10) | |
| Q9 | (a) Find the Laurent's series of $f(z) = \frac{z}{(z^2+1)(z^2+4)}$ in $1 < \|z\| < 2$.<br>(b) Evaluate $\int_C \frac{z-1}{(z+1)^2(z-2)}dz$, where $C$ is $\|z - i\| = 2$. | (5*2=10) | |